



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2010

PRELIMINARY SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 1

Mathematics Extension I

TIME ALLOWED: 1 HOUR

PLUS 5 MINUTES READING TIME

Outcomes Assessed	Questions	Marks
Demonstrates the ability to manipulate and simplify numeric and algebraic expressions and solves problems involving equations	1	
Solves problems involving inequalities, indices and logs	2	
Uses appropriate techniques to solve problems involving plane and circle geometry	3	

Question	1	2	3	Total	%
Marks	/12	/16	/12	/40	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each Question is to be started in a new booklet.

Question 1. (12 marks)

- a) i) Factorise $y^2 + 4y + 4$ **1**
- ii) Factorise $y^3 + 8$ **1**
- iii) Hence, simplify $\frac{y^3 + 8}{3y - 6} \times \frac{y^2 - 4}{y^2 + 4y + 4}$ **1**
- b) Express $1.\dot{3}\dot{8}$ as a fraction in its simplest form. **2**
- c) Solve $3x^2 - 5x - 3 = 0$, leaving you answer in exact form **2**
- d) Show that $\frac{1}{3 - \sqrt{2}} + \frac{1}{3 + \sqrt{2}}$ is a rational number. **2**
- f) Solve for u , v and w . **3**
- $$\begin{array}{rclcl} 3u & + & v & - & 4w & = & -4 \\ u & - & 2v & + & 7w & = & -7 \\ 4u & + & 3v & - & w & = & 9 \end{array}$$

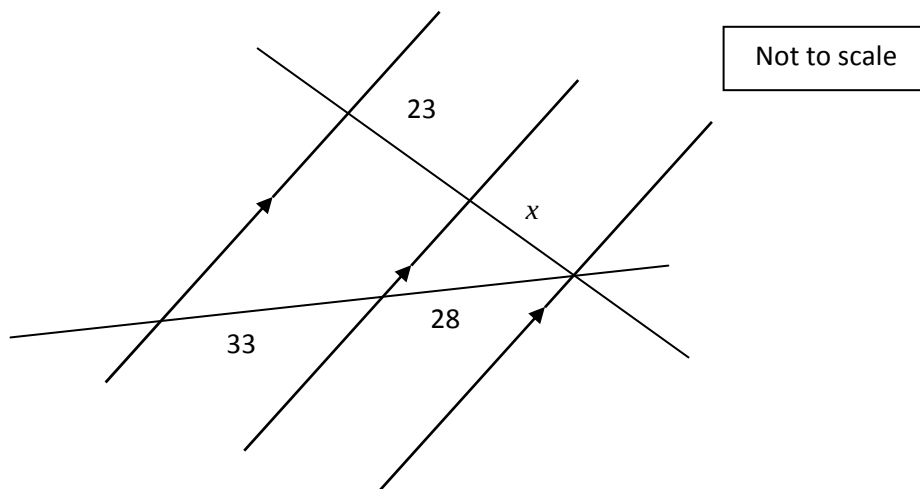
Question 2. *Start a separate booklet* (16 marks)

- a) Solve the inequality $\frac{2x+1}{x+4} \geq 1$ **3**
- b) i) sketch the graph of $y = |2x - 2|$ **2**
- ii) hence or otherwise solve $|2x - 2| \leq |x - 3|$ **2**
- c) Simplify $\log_2 64 - \log_2 8$ **2**
- d) If $\log_5 7 = 1.21$ and $\log_5 2 = 0.43$, evaluate $\log_5 98$ **2**
- e) Solve $3^{x-5} = 238$ correct to three significant figures **2**
- f) Solve $4^x - 9(2^x) + 8 = 0$ **3**

Question 3. *Start a separate booklet* (12 marks)

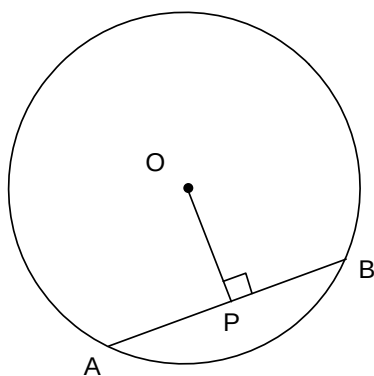
- a) Find the value of x in the following diagram

2



- b) Prove the perpendicular from the centre of a circle to a chord bisects the chord.

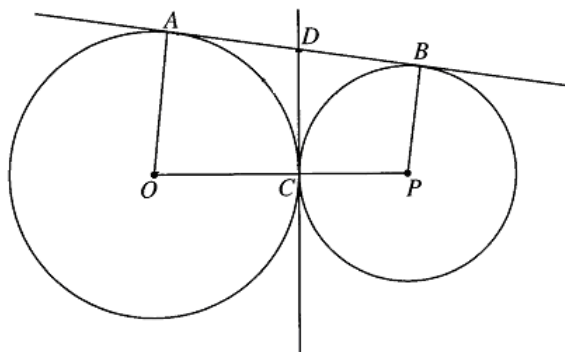
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Question 3 continues on the next page.

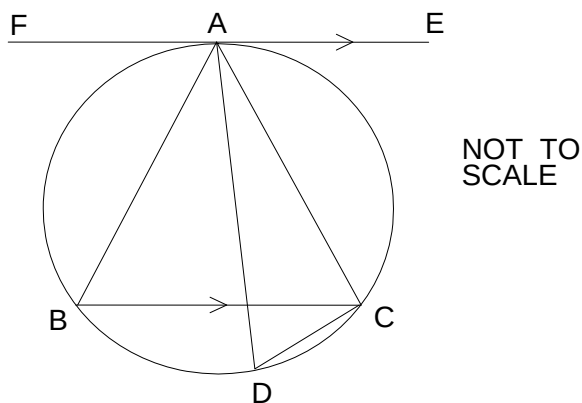
- c) Two circles, centres O and P , intersect at point C only.

AB is a common tangent to the two circles which meets the tangent through C at D .



- (i) Prove that $DA = DB$ 2
- (ii) Prove that quadrilateral $AOCD$ is cyclic. 2

d)



In the diagram the points A , B , C and D lie on the circle, FAE is a tangent that touches the circle at A . FE is parallel to BC .

Let $\angle FAB = \alpha$.

- (i) Explain why $\angle ACB = \alpha$ 1
- (ii) Hence prove that $\angle ACB = \angle ADC$ 2



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Solutions

Question 1. (12 marks)

- a) i) Factorise $y^2 + 4y + 4$

Solution

$$y^2 + 4y + 4 = (y + 2)^2$$

Marking guideline: 1 mark for correct response

- ii) Factorise $y^3 + 8$

Solution

$$y^3 + 8 = (y + 2)(y^2 - 2y + 4)$$

Marking guideline: 1 mark for correct response

- iii) Hence, simplify $\frac{y^3 + 8}{3y - 6} \times \frac{y^2 - 4}{y^2 + 4y + 4}$

Solution

$$\begin{aligned} \frac{y^3 + 8}{3y - 6} \times \frac{y^2 - 4}{y^2 + 4y + 4} &= \frac{(y + 2)(y^2 - 2y + 4)}{3(y - 2)} \times \frac{(y + 2)(y - 2)}{(y + 2)^2} \\ &= \frac{(y^2 - 2y + 4)}{3} \times \frac{1}{1} \\ &= \frac{(y^2 - 2y + 4)}{3} \end{aligned}$$

Marking guideline: 1 mark for correct response

b) Express $1.\dot{3}8$ as a fraction in its simplest form.

2

Solution

$$\begin{aligned}x &= 1.3888\dots \\10x &= 13.888\dots \\100x &= 138.888\dots\end{aligned}$$

$$100x - 10x = 125$$

$$90x = 125$$

$$x = \frac{125}{90}$$

$$= \frac{25}{18} \text{ or } 1\frac{7}{8}$$

Marking guideline:

2 marks for correct response OR

1 mark for correct procedure but with arithmetic error

c) Solve $3x^2 - 5x - 3 = 0$, leaving your answer in exact form

2

Solution

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{5 \pm \sqrt{(-5)^2 - 4(3)(-3)}}{2 \cdot 3}$$

$$= \frac{5 \pm \sqrt{61}}{6}$$

Marking guideline:

2 marks for correct response OR

1 mark for correct procedure but with arithmetic error OR

1 mark for not answering in *exact* form.

d) Show that $\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}}$ is a rational number.

2

Solution

$$\begin{aligned}\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}} &= \frac{1}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} + \frac{1}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}} \\ &= \frac{3+\sqrt{2}}{9-2} + \frac{3-\sqrt{2}}{9-2} \\ &= \frac{3+\sqrt{2} + 3-\sqrt{2}}{7} \\ &= \frac{6}{7}\end{aligned}$$

Which is a rational number.

Marking guideline:

2 marks for correct response OR

1 mark for correct procedure but with arithmetic error

e) Solve for u , v and w .

$$\begin{aligned}3u + v - 4w &= -4 \quad (1) \\ u - 2v + 7w &= -7 \quad (2) \\ 4u + 3v - w &= 9 \quad (3)\end{aligned}$$

Solution

$$(2) \times 3 \quad 3u - 6v + 21w = -21 \quad (4)$$

$$(2) \times 4 \quad 4u - 8v + 28w = -28 \quad (5)$$

$$(1) - (4) \quad 7v - 25w = 17 \quad (6)$$

$$(3) - (5) \quad 11v - 29w = 37 \quad (7)$$

$$(6) \times 11 \quad 77v - 275w = 187 \quad (8)$$

$$(7) \times 7 \quad 77v - 203w = 259 \quad (9)$$

$$(9) - (8) \quad 72w = 72$$

$$w = 1$$

Substituting $w = 1$ into the original equations

$$3u + v = 0 \quad (1A)$$

$$u - 2v = -14 \quad (2A)$$

$$(2) \times 3 \quad 3u - 6v = -42 \quad (10)$$

$$(1) - (10) \quad 7v = 42$$

$$v = 6$$

Substituting $w = 1$, $v = 6$ into (1A)

$$3u + 6 = 0$$

$$3u = -6$$

$$u = -2$$

$$\therefore u = -2, \quad v = 6, \quad w = 1$$

Marking guideline:

3 marks for correct response OR

2 marks for correct procedure but with one arithmetic error OR

1 mark for correct procedure with 2 or more arithmetic errors

Question 2. Start a separate booklet (16 marks)

a) Solve the inequality $\frac{2x+1}{x+4} \geq 1$

Solution

NOTE: $x \neq -4$

$$\frac{2x+1}{x+4} \geq 1$$

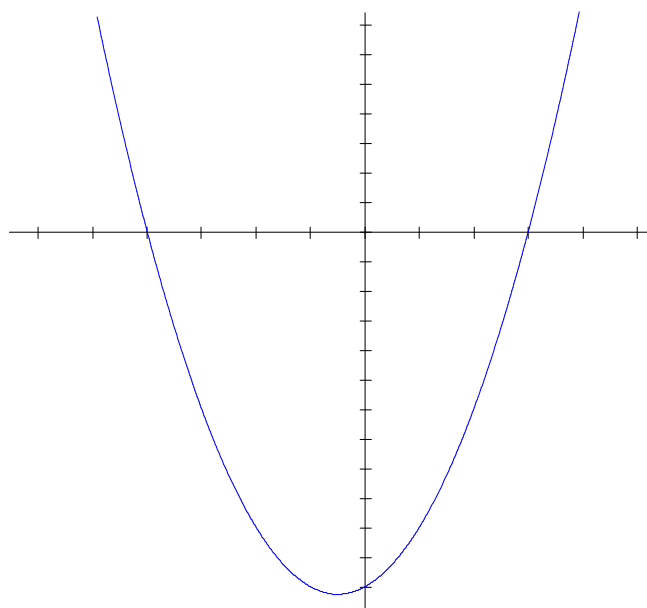
$$\frac{(2x+1)(x+4)^2}{x+4} \geq (x+4)^2$$

$$(2x+1)(x+4) - (x+4)^2 \geq 0$$

$$(x+4)((2x+1) - (x+4)) \geq 0$$

$$(x+4)(x-3) \geq 0$$

$$\therefore x < -4 \quad \text{or} \quad x \geq 3$$



Marking guideline:

3 marks for correct response OR

2 marks for correct procedure but with one arithmetic error OR

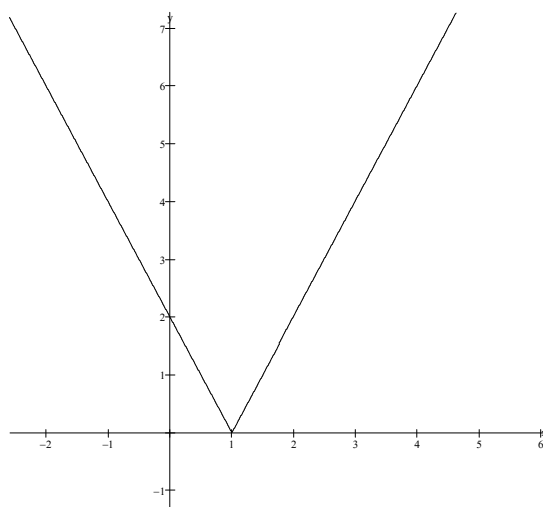
2 marks for $x \leq -4$ in the solution and not the correct $x < -4$

1 mark for $x > -4$ or $x > 3$ (i.e. neglects to graph or similar)

b) i) sketch the graph of $y = |2x - 2|$

2

Solution



Marking guideline:

2 marks for correct response OR

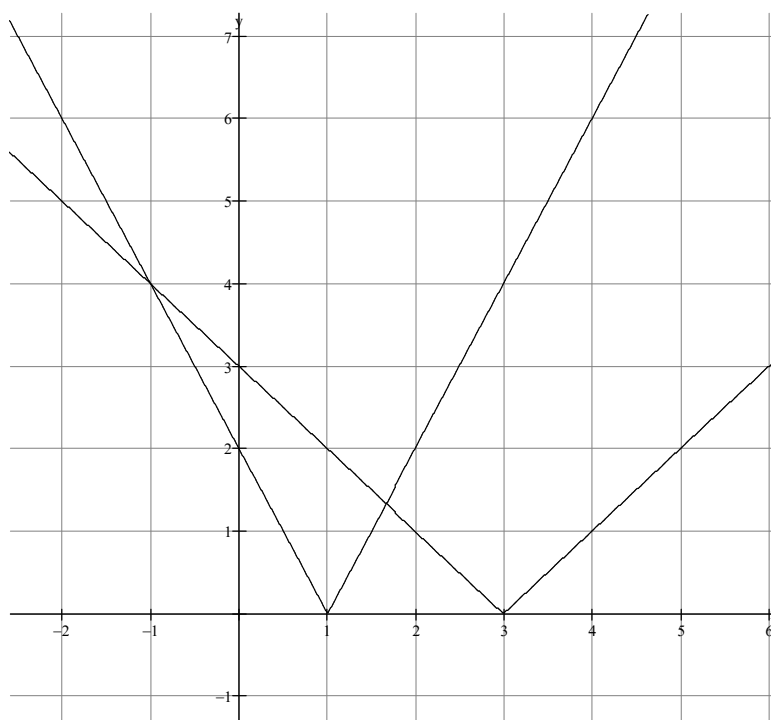
1 mark for correct shape with errors

ii) hence or otherwise solve $|2x - 2| \leq |x - 3|$

Solution

Graph

$$y = |x - 3|$$



From the graph, the intersections occur when both branches of $y = |2x - 2|$ cuts the negative branch of $y = |x - 3|$.

That is, $y = |2x - 2|$ intersects the left branch of $y = |x - 3|$ i.e. the line $y = 3 - x$

$$(2x - 2) \leq -(x - 3)$$

$$2x - 2 \leq -x + 3$$

$$3x \leq 5$$

$$x \leq \frac{5}{3}$$

And, the left branch of $y = |2x - 2|$ i.e. the line $y = -2x + 2$ intersects the left branch of $y = |x - 3|$ i.e. the line $y = 3 - x$

$$-(2x - 2) \leq -(x - 3)$$

$$-2x + 2 \leq -x + 3$$

$$-x \leq 1$$

$$x \geq -1$$

So solution is $-1 \leq x \leq \frac{5}{3}$

Marking guideline: 2 marks for correct response OR

1 mark for partial solution

c) Simplify $\log_2 64 - \log_2 8$

Solution

$$\log_2 64 - \log_2 8 = \log_2 \frac{64}{8}$$

$$= \log_2 8$$

$$= \log_2 2^3$$

$$= 3 \log_2 2$$

$$= 3$$

Marking guideline:

2 marks for correct response OR

1 mark for evidence of two logs laws

d) If $\log_5 7 = 1.21$ and $\log_5 2 = 0.43$, evaluate $\log_5 98$

Solution

$$\log_5 98 = \log_5 (49 \times 2)$$

$$= \log_5 49 + \log_5 2$$

$$= \log_5 7^2 + \log_5 2$$

$$= 2 \log_5 7 + \log_5 2$$

$$= 2 \times 1.21 + 0.43$$

$$= 2.85$$

Marking guideline:

2 marks for correct response OR

1 mark for evidence of two logs laws

e) Solve $3^{x-5} = 238$ correct to three significant figures

Solution

$$3^{x-5} = 238$$

$$\log_{10} 3^{x-5} = \log_{10} 238$$

$$(x-5)\log_{10} 3 = \log_{10} 238$$

$$x-5 = \frac{\log_{10} 238}{\log_{10} 3}$$

$$x = 5 + \frac{\log_{10} 238}{\log_{10} 3}$$

$$= 9.9810754\dots$$

$$= 9.98 \quad \text{Correct to 3 sig. figs.}$$

Marking guideline:

2 marks for correct response OR

1 mark for not correcting to 3 sig. figs.

f) Solve $4^x - 9(2^x) + 8 = 0$

Solution

Let $m = 2^x$

$$4^x - 9(2^x) + 8 = 0$$

$$(2^x)^2 - 9(2^x) + 8 = 0$$

$$m^2 - 9m + 8 = 0$$

$$(m-8)(m-1) = 0$$

$$m = 8 \text{ or } 1$$

But $m = 2^x$, so

$$2^x = 8$$

$$2^x = 1$$

OR

$$x = 3$$

$$x = 0$$

Marking guideline:

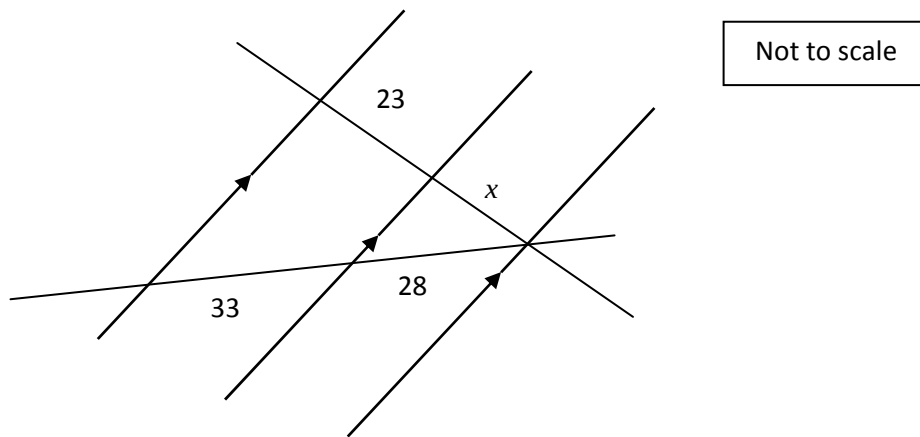
3 marks for correct response OR

2 marks for correct procedure but with one arithmetic error OR

1 mark for finding $m = 8$ or 1

Question 3. *Start a separate booklet* (12 marks)

a) Find the value of x in the following diagram



Solution

$$\frac{x}{23} = \frac{28}{33} \quad \text{Transversal cuts off intercepts in the same ratio}$$

$$x = \frac{644}{33}$$

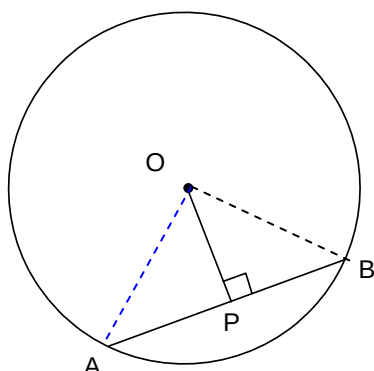
$$= 19\frac{17}{33}$$

Marking guideline:

2 marks for correct response OR

1 mark correct procedure with one arithmetic error

- b) Prove the perpendicular from the centre of a circle to a chord bisects the chord.



Solution

Construction: Join OA and OB

In $\triangle AOP$ and $\triangle BOP$

$$\angle APO = \angle BPO$$

Given – perpendicular line OP & AB

OP

Common

$$OA = OB$$

Equal radii

$$\therefore \triangle AOP \cong \triangle BOP$$

RHS

So, $AP = BP$

Corresponding sides of congruent triangles.

$\therefore AP$ bisects AB

Marking guideline:

3 marks for correct response OR

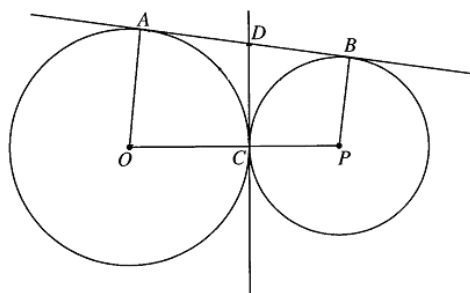
2 marks for correct procedure with one missing reason OR

2 marks for proving congruent triangles but not stating consequence OR

1 mark for correct procedure but with two or more missing reasons.

- c) Two circles, centres O and P , intersect at point C only.

AB is a common tangent to the two circles which meets the tangent through C at D .



- (i) Prove that $DA = DB$

Solution

$$DA = DC \quad [\text{tangents from an external point are equal}]$$

$$DB = DC \quad [\text{tangents from an external point are equal}]$$

$$\therefore DA = DB \quad [\text{both intervals are equivalent to } DC]$$

Marking guideline:

2 marks for correct response OR

1 mark for incorrect or no reasoning

- (ii) Prove that quadrilateral $AOCD$ is cyclic.

2

Solution

$$\angle OAD = 90^\circ \quad [\text{radius meets a tangent at right angles}]$$

$$\angle OCD = 90^\circ \quad [\text{radius meets a tangent at right angles}]$$

$$= 180^\circ$$

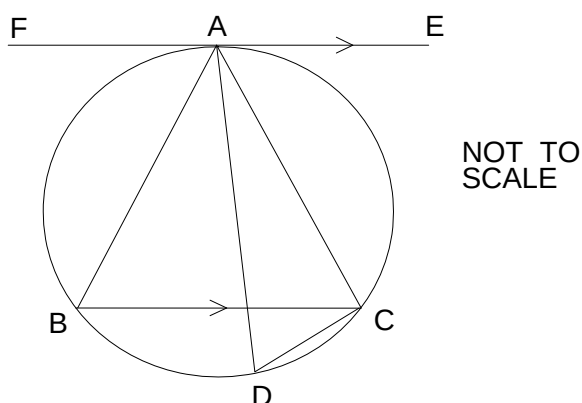
$\therefore AOCD$ is a cyclic quadrilateral [opposite angles supplementary]

Marking guideline:

2 marks for correct response OR

1 mark for lack of reasons

d)



In the diagram the points A, B, C and D lie on the circle, FAE is a tangent that touches the circle at A. FE is parallel to BC.

Let $\angle FAB = \alpha$.

- (i) Explain why $\angle ACB = \alpha$

Solution

The angle between a chord and a tangent is equal to the angle subtended by the chord in the alternate segment.

Marking guideline: 1 mark for correct response

- (ii) Hence prove that $\angle ACB = \angle ADC$

Solution

$FAB = ABC = \alpha$ [Alternate angles of parallel lines]

$ABC = ADC = \alpha$ [Angles standing on the same arc subtended by the same chord are equal]

And $\angle ACB = \alpha$ [From above]

So $\angle ACB = \angle ADC$

Marking guideline: 2 marks for correct response OR

1 mark for lack of reasons